Industrial Vision

Lecture 17 - Structure from Motion I

Pose from E, triangulation, and chirality

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Where we are

Last lecture we **estimated** F from matched points. The **eight/seven-point** algorithm gave a small solver with the help of **RANSAC**. We then got depth from $Z = fb/d$, but **only because we were told the baseline** b.

Today nobody tells us the camera motion. Think of walking around a building with a phone. We must find both of these from the images alone:

$$\underbrace{(R_i, \mathbf{t}_i)}_{\text{motion: where the camera was}} \qquad \text{and} \qquad \underbrace{\{\mathbf{X}_j\}}_{\text{structure: where the 3-D points are}}$$

This is called **structure from motion** (SfM). Today we do the **two-view** case in four steps: get E, split E into a pose, triangulate points, and pick the one pose that makes physical sense.

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We can never get the absolute size

When the baseline was **known**, disparity gave depth in real units. When the motion is **unknown**, two different worlds give the same pictures.

A **big scene far away, with a big camera move**, and a **small scene close by, with a small camera move**, give exactly the same images.

$$\text{disparity} \sim \frac{f \, \|\mathbf{t}\|}{Z}$$

Only the **ratio** $\|\mathbf{t}\| / Z$ shows up in the image. Double $\mathbf{t}$ and double every depth, and nothing in the image changes.

So SfM gives the scene **up to one unknown scale factor**. The shape is right, but the units are not: a depth of "2.0" does not mean 2 metres.

One extra measurement fixes it: a known baseline, one known distance in the scene, or an IMU reading. Without any of these, the scale stays unknown.

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Step 1: the essential matrix from F

The camera is **calibrated**, so K is known. Last time we saw $F = K_R^{-\top}\,[\mathbf{t}]_\times R\,K_L^{-1}$. Multiply by $K_R^{\top}$ on the left and K_L on the right to remove the intrinsics:

$$E = K_R^{\top}\,F\,K_L = [\mathbf{t}]_\times R.$$

E has a special shape. A true essential matrix always has singular values $(\sigma, \sigma, 0)$: **two equal, one zero**.

Noise breaks this shape. The F we estimated is noisy, so the E we get from it will have three different singular values $\sigma_1' \geq \sigma_2' \geq \sigma_3'$. We force the shape back: take the SVD and replace the two largest by their average and the smallest by zero.

$$E = U\,\mathrm{diag}(\sigma_1', \sigma_2', \sigma_3')V^{\top} \quad \longrightarrow \quad \hat{E} = U\,\mathrm{diag}(\sigma, \sigma, 0)V^{\top}, \qquad \sigma = \tfrac{\sigma_1' + \sigma_2'}{2}.$$

This $\hat{E}$ is the closest valid essential matrix to E (in the Frobenius norm). Since E is only known up to scale anyway, we may also set $\sigma = 1$ and use $U\,\mathrm{diag}(1, 1, 0)V^{\top}$. In practice this just means keeping U and V and discarding the singular values. Because $\sigma = \|\mathbf{t}\|$, it amounts to choosing the baseline length as the unit of the reconstruction.

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Step 2a: split E into "skew-symmetric × rotation"

We have E and want R and $\mathbf{t}$ back. **Goal:** write $E = S\,R$ with $S = [\mathbf{t}]_\times$ skew-symmetric and R a rotation. Start from $E = U\,\mathrm{diag}(1,1,0)V^\top$ (scale does not matter) and define two helper matrices:

$$Z = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (Z^\top = -Z),$$

$$W = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (WW^\top = I,\ \det W = 1).$$

Z is skew-symmetric. W is a rotation by 90° about the z-axis.

Multiply them out and check:

$$ZW = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad ZW^\top = -\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Choice 1. Replace $\mathrm{diag}(1,1,0)$ by ZW and slip $U^\top U = I$ into the middle:

$$E = U\,Z\,W\,V^\top = \underbrace{(UZU^\top)}_{S_1}\,\underbrace{(UWV^\top)}_{R_1}.$$

Choice 2. Use $ZW^\top = -\,\mathrm{diag}(1,1,0)$ instead:

$$E = -\,U\,Z\,W^\top V^\top = \underbrace{(-UZU^\top)}_{S_2}\,\underbrace{(UW^\top V^\top)}_{R_2}.$$

Check that each piece is the right kind of matrix: $S^\top = UZ^\top U^\top = -S$, so S is **skew-symmetric**; and R is a product of orthogonal matrices, so it is **orthogonal**. Hence there are **two possible rotations**, $R_1 = UWV^\top$ and $R_2 = UW^\top V^\top$.

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Step 2b: find $\mathbf{t}$, and count the candidates

The translation lives in the null space of S. Since $S = [\mathbf{t}]_\times$,

$$S\,\mathbf{t} = \mathbf{t} \times \mathbf{t} = \mathbf{0}.$$

So $\mathbf{t}$ is the vector that S sends to zero. Try $\mathbf{u}_3$, the **third column of U**. Because $U^\top \mathbf{u}_3 = (0,0,1)^\top$,

$$S\,\mathbf{u}_3 = U Z U^\top \mathbf{u}_3 = U Z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = U \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}.$$

So $\mathbf{t} = \lambda\,\mathbf{u}_3$ for some $\lambda \neq 0$.

- The **size** of λ cannot be found. This is the scale ambiguity from before.
- The **sign** of λ does matter: $+\mathbf{u}_3$ and $-\mathbf{u}_3$ put the second camera on opposite sides.

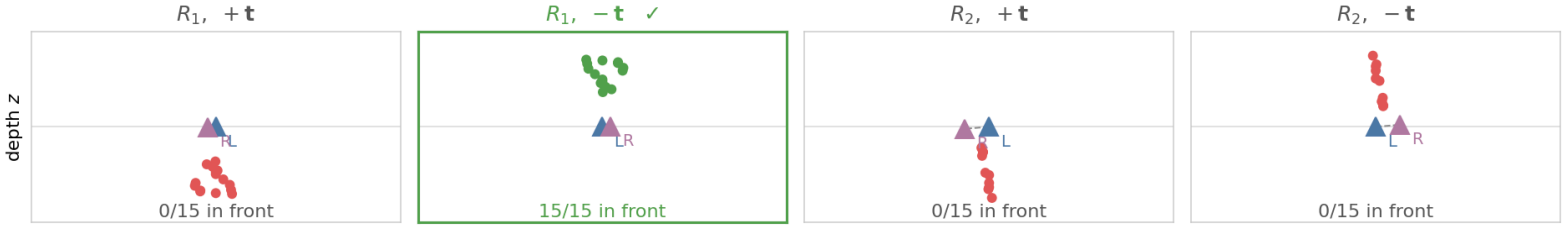
Four candidates. Two rotations times two signs of $\mathbf{t}$ gives four possible right cameras (with $P_L = [\,I \mid \mathbf{0}\,]$):

$$P_R = [\,UWV^\top \mid +\mathbf{u}_3\,] \quad \text{or} \quad [\,UWV^\top \mid -\mathbf{u}_3\,]$$
$$[\,UW^\top V^\top \mid +\mathbf{u}_3\,] \quad \text{or} \quad [\,UW^\top V^\top \mid -\mathbf{u}_3\,]$$

All four give the **same** E (up to sign), so the images alone cannot tell them apart. We need one extra physical rule to pick the right one. That rule is **chirality** (Step 4): the 3-D points must be **in front of** both cameras.

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Step 2c: the four candidates, and the det $R = +1$ check



Top view of the four candidates. Green points are in front of both cameras, red points are behind at least one. Only one candidate is all green.

Make sure $\det R = +1$. $UWV^\top$ is always orthogonal, but the SVD may return U or V with determinant -1. Then $\det R = -1$, which is a **mirror reflection**, not a rotation. No real camera motion can mirror the world. **Fix:** if $\det U < 0$, flip the sign of the third column of U; if $\det V < 0$, flip the third column of V. This does **not change** E, because that column is multiplied by the singular value 0.

Why exactly four? Two independent yes/no choices: which rotation (R_1 or R_2), and which sign of $\mathbf{t}$. The two rotations differ by a 180° **turn about the baseline**, so they are called the **twisted pair**. In the picture: one candidate has the points in front of both cameras, one has them behind both, and the other two have them in front of one camera and behind the other.

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Step 3: triangulation by the direct linear transform

Now we have a candidate pose $(R, \mathbf{t})$ from Step 2. With the world frame at the left camera, the two cameras are $P = K_L[\,I \mid \mathbf{0}\,]$ and $P' = K_R[\,R \mid \mathbf{t}\,]$. For each matched pair $\mathbf{x} \leftrightarrow \mathbf{x}'$ we want the 3-D point $\mathbf{X}$. In each view the image point and the projection of $\mathbf{X}$ are the **same point up to scale** ($\tilde{\mathbf{x}} \sim P\tilde{\mathbf{X}}$ and $\tilde{\mathbf{x}}' \sim P'\tilde{\mathbf{X}}$), so they are parallel vectors and their cross product is zero:

$$\begin{aligned} \text{left view:} \quad & \tilde{\mathbf{x}} \times (P\tilde{\mathbf{X}}) = \mathbf{0} \iff [\tilde{\mathbf{x}}]_\times P\,\tilde{\mathbf{X}} = \mathbf{0}, \\ \text{right view:} \quad & \tilde{\mathbf{x}}' \times (P'\tilde{\mathbf{X}}) = \mathbf{0} \iff [\tilde{\mathbf{x}}']_\times P'\,\tilde{\mathbf{X}} = \mathbf{0}. \end{aligned}$$

Expand the left view. Write $\mathbf{p}_i^\top$ for the i-th **row** of P, so $P\tilde{\mathbf{X}} = (\mathbf{p}_1^\top\tilde{\mathbf{X}},\ \mathbf{p}_2^\top\tilde{\mathbf{X}},\ \mathbf{p}_3^\top\tilde{\mathbf{X}})^\top$. With $\tilde{\mathbf{x}} = (x, y, 1)^\top$, the cross product is

$$\tilde{\mathbf{x}} \times P\tilde{\mathbf{X}} = \begin{bmatrix} y\,\mathbf{p}_3^\top\tilde{\mathbf{X}} - \mathbf{p}_2^\top\tilde{\mathbf{X}} \\ \mathbf{p}_1^\top\tilde{\mathbf{X}} - x\,\mathbf{p}_3^\top\tilde{\mathbf{X}} \\ x\,\mathbf{p}_2^\top\tilde{\mathbf{X}} - y\,\mathbf{p}_1^\top\tilde{\mathbf{X}} \end{bmatrix} = \mathbf{0}.$$

Each row is **linear** in $\tilde{\mathbf{X}}$. The third row is $-(x \cdot \text{row } 1 + y \cdot \text{row } 2)$, so only **two** are independent. Keep the first two (the second with its sign flipped):

$$(x\,\mathbf{p}_3^\top - \mathbf{p}_1^\top)\,\tilde{\mathbf{X}} = 0, \qquad (y\,\mathbf{p}_3^\top - \mathbf{p}_2^\top)\,\tilde{\mathbf{X}} = 0.$$

The right view is the same with x', y' and the rows $\mathbf{p}_i'^\top$ of P'. Stack the two equations from **each** view (rows 1-2 from the left view, rows 3-4 from the right):

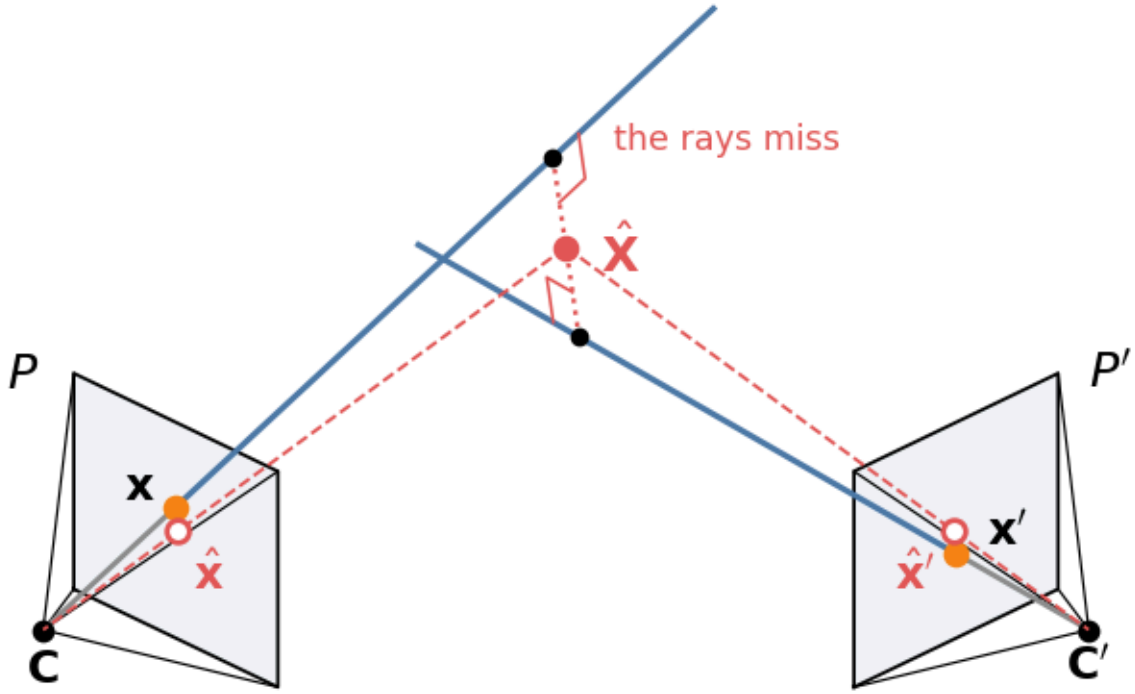
$$A\,\tilde{\mathbf{X}} = \mathbf{0}, \qquad A = \begin{bmatrix} x\,\mathbf{p}_3^\top - \mathbf{p}_1^\top \\ y\,\mathbf{p}_3^\top - \mathbf{p}_2^\top \\ x'\mathbf{p}_3'^\top - \mathbf{p}_1'^\top \\ y'\mathbf{p}_3'^\top - \mathbf{p}_2'^\top \end{bmatrix}$$

Solve it like every DLT so far. A is 4×4 and $\tilde{\mathbf{X}}$ has 4 unknowns up to scale.

- **No noise:** the two rays meet exactly, A has **rank 3**, and $\tilde{\mathbf{X}}$ is its null vector.
- **With noise:** the two rays just miss each other, A has full rank, and we take the **right singular vector of the smallest singular value**.

Finally divide by the fourth coordinate to get (X, Y, Z).

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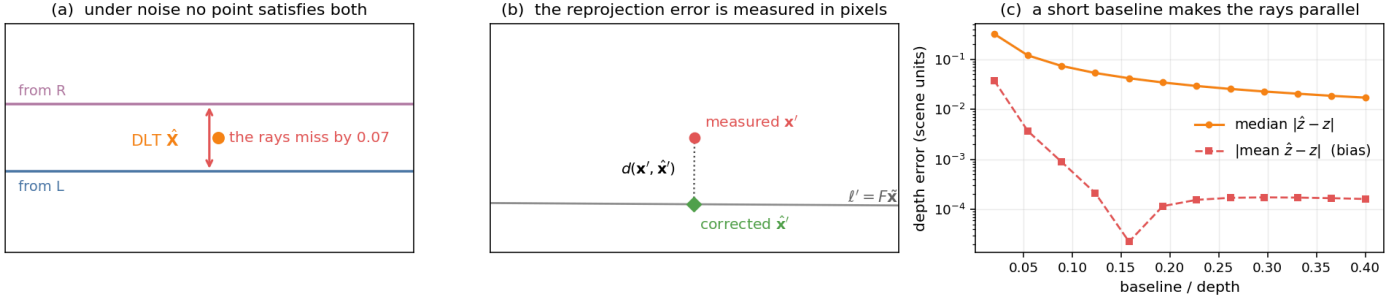
With noise, the rays from $\mathbf{C}$ through $\mathbf{x}$ and from $\mathbf{C}'$ through $\mathbf{x}'$ just miss each other; $\hat{\mathbf{X}}$ is the point we report. Projecting $\hat{\mathbf{X}}$ back with each camera (dashed lines) gives $\hat{\mathbf{x}} = \pi(P, \hat{\mathbf{X}})$ and $\hat{\mathbf{x}}' = \pi(P', \hat{\mathbf{X}})$. These land close to the observed $\mathbf{x}, \mathbf{x}'$, but not exactly on them; that gap is the reprojection error.

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What did triangulation actually minimize?

The DLT minimizes $\|A\tilde{\mathbf{X}}\|$. This is an **algebraic** number, **not a distance** in the image. What we really care about is the **reprojection error** in pixels: how far the point lands from where we saw it.

$$E_{\text{reproj}}(\hat{\mathbf{X}}) = d\big(\mathbf{x},\ \pi(P,\hat{\mathbf{X}})\big)^2 + d\big(\mathbf{x}',\ \pi(P',\hat{\mathbf{X}})\big)^2.$$

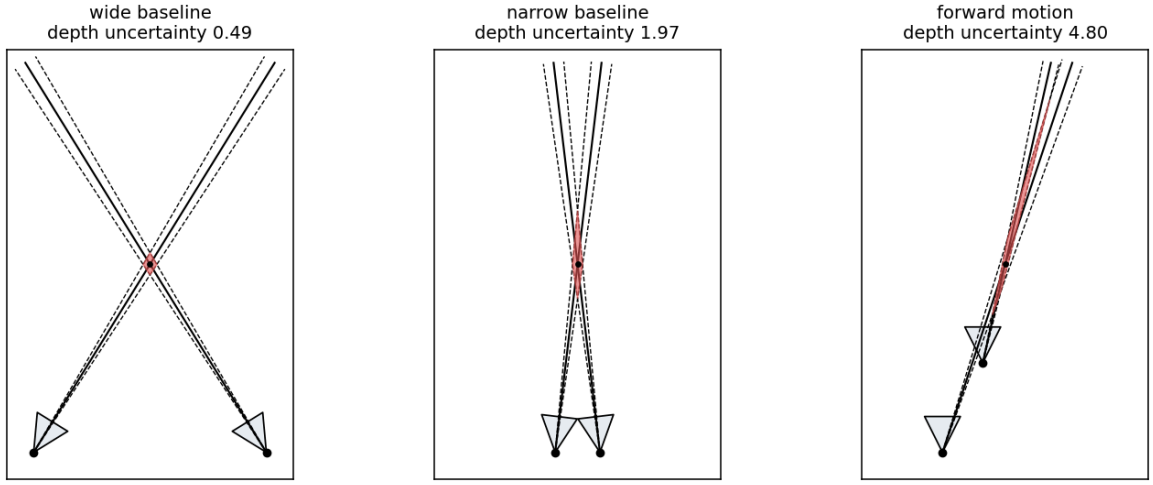


How to do better. Minimizing E_{reproj} is called **optimal triangulation**. In practice we start from the DLT answer and improve it with **Levenberg-Marquardt**. Improving all the points **and** all the poses together is **bundle adjustment**.

Short baselines are bad. When the baseline is small compared with the depth, the two rays are almost parallel. A tiny pixel error then moves the meeting point a lot along the depth direction, and the depth estimate is also **biased**. This is why a **wide baseline** gives better 3-D.

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Triangulation uncertainty depends on the camera pose



Top view. Each ray (solid) is widened by the pixel noise into a wedge (dashed). The red region is where the two wedges overlap: every point in it fits both observations. The noise is the same in all three panels.

- **Uncertainty grows as the rays become more parallel.**
- **Forward motion is the worst case.**
- **The tradeoff.** Nearby views look alike, so **matching is easy**, but **triangulation is poor**. Distant views triangulate well but are **harder to match**.

Step 4: chirality picks the real pose

We have four candidates, but **only one can be real**: a point that a camera sees must be **in front of** that camera (positive depth). The other three put points behind at least one camera.

The test. For each of the four candidates $(R, \mathbf{t})$:

1. triangulate the matched points (Step 3);
2. check the depth Z of each point in **both** cameras;
3. keep the candidate where the points have $Z > 0$ in both.

This fixes the **sign** of $\mathbf{t}$. Its **size** is still unknown.

With noise, a few points may land on the wrong side. So we count, and pick the candidate with the **most** points in front.

We only need a **few** matches for this. We are choosing between four options, not building a depth map.

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When K is unknown: cameras straight from F

Without K we cannot form E. But we can still get **some** pair of cameras that agrees with F, and that is enough to triangulate.

The recipe. Fix the left camera and build the right one from F:

$$P_L = [\, I \mid \mathbf{0} \,], \qquad P_R = \left[\, [\mathbf{e}']_\times F \;\middle|\; \mathbf{e}' \,\right],$$

where $\mathbf{e}'$ is the epipole in the right image, found as the null vector of $F^\top$ (so $\mathbf{e}'^\top F = \mathbf{0}^\top$).

Then triangulate exactly as in Step 3. There is **no chirality step** and **no four-way choice**: we are not trying to find the true pose, only *a* pose consistent with F.

Check that it gives back F. For cameras $[I \mid \mathbf{0}]$ and $[M \mid \mathbf{m}]$ the fundamental matrix is $[\mathbf{m}]_\times M$. Here $\mathbf{m} = \mathbf{e}'$ and $M = [\mathbf{e}']_\times F$, so

$$[\mathbf{e}']_\times [\mathbf{e}']_\times F = (\mathbf{e}'\mathbf{e}'^\top - \|\mathbf{e}'\|^2 I)\, F = -\|\mathbf{e}'\|^2 F,$$

using $\mathbf{e}'^\top F = \mathbf{0}^\top$. That is F up to scale.

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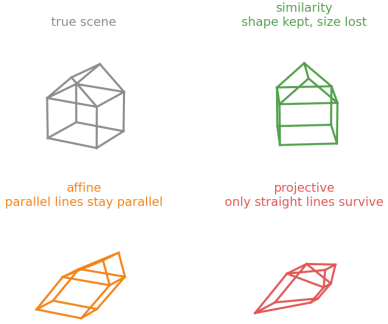
Without K, the 3-D shape is only known up to a projective warp

Why. Take any invertible 4×4 matrix H. Then

$$\mathbf{x} = P\,\mathbf{X} = (PH^{-1})\,(H\mathbf{X}).$$

The new cameras PH^{-1} and the new points $H\mathbf{X}$ give **exactly the same pixels**. The images cannot tell the two worlds apart.

- K **known:** H must keep the cameras calibrated. Only a rotation, a translation and a scale survive: a **similarity** (7 numbers).
- **Nothing known: any** H works: a **projective** transform (15 numbers).



What we know	Recovered up to	What stays true
nothing	**projective** (15 dof)	straight lines stay straight
+ plane at infinity	**affine** (12 dof)	+ parallel lines stay parallel
+ K (angles)	**similarity** (7 dof)	+ angles and length ratios
+ one real length	**Euclidean** (6 dof)	+ true sizes

Climbing this ladder is **stratified reconstruction**: add facts about the scene (parallel lines, right angles, a known length) or the camera (calibration, an IMU).

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Questions

Q. Is every orthogonal matrix a rotation?

A. No. $A^\top A = I$ only gives $\det A = \pm 1$. A real rotation needs $\det A = +1$. With $\det A = -1$ it is a **mirror reflection**. That is why Step 2c checks $\det R = +1$.

Q. Triangulation finds points from known poses, PnP finds a pose from known points. Can we alternate them?

A. Yes. That is the **incremental SfM** loop: triangulate, add the next camera by PnP, triangulate more. It works, but errors **pile up (drift)**, because each step trusts the other's output as exact. Solving for everything **together** is **bundle adjustment**

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Summary

- **SfM** finds the camera motion $(R_i, \mathbf{t}_i)$ **and** the 3-D points $\{\mathbf{X}_j\}$ from images alone. The result is only known **up to scale**, because only $\|\mathbf{t}\|/Z$ shows up in the images.
- **Step 1.** $E = K_R^\top F K_L = [\mathbf{t}]_\times R$. Its singular values are $(\sigma, \sigma, 0)$, because R keeps lengths and "cross with $\mathbf{t}$" kills $\mathbf{t}$ and stretches the perpendicular plane evenly. Noise breaks this, so we average the two largest and zero the smallest.
- **Step 2.** With $ZW = \mathrm{diag}(1, 1, 0)$ we get $E = (UZU^\top)(UWV^\top)$, so $R \in \{UWV^\top, UW^\top V^\top\}$, and $\mathbf{t} = \pm\mathbf{u}_3$ (null space of S). That is **four candidates**. Always make sure $\det R = +1$.
- **Step 3.** Triangulate with the DLT: stack $[\tilde{\mathbf{x}}]_\times P\tilde{\mathbf{X}} = \mathbf{0}$ from both views into a 4×4 system and take the last singular vector. This minimizes an **algebraic** error, not pixels; refine with Levenberg-Marquardt (L22). Short baselines give poor, biased depth.
- **Step 4. Chirality**: keep the candidate that puts the points **in front of both** cameras.
- **No K:** use $P_L = [I \mid \mathbf{0}]$, $P_R = [[\mathbf{e}']_\times F \mid \mathbf{e}']$. The shape is then known only up to a **projective** warp; extra facts climb the ladder to affine, similarity and Euclidean.

Next lecture: from two views to many. Today's two-view result becomes the **seed**. Each new image is placed by **PnP** against the points we already have, new points are triangulated as today, and everything is polished by **bundle adjustment**, which minimizes the reprojection error over all cameras and points at once.

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References

1. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge Univ. Press, 2004. See **Ch. 9**, in particular §9.6 on recovering E and the fourfold ambiguity, **Ch. 12**, "Structure Computation", §12.2 on the linear triangulation methods and **§12.5** on optimal triangulation by a polynomial of degree six, due to Hartley and Sturm (1997), and §9.5 and Ch. 10 on projective reconstruction and the stratified ladder.

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